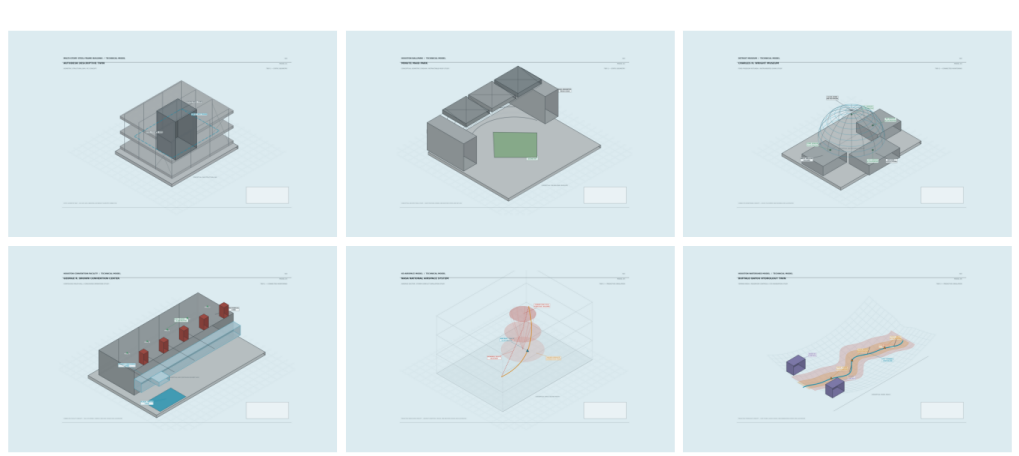
ITAI 4371

# DIGITAL TWIN
# VERISIMILITUDE

Definitions at Three Levels




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**Representative → Monitoring → Predictive**

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# Digital Twin Definitions at Three Levels of Verisimilitude

A digital twin is a virtual representation of something that exists in the physical world. Depending on its level of verisimilitude, a digital twin can represent the physical object or environment, monitor its current conditions, or predict how it may behave in the future.

The progression can be understood through three layers:

**Representative Layer** → **Monitoring Layer** → **Predictive Layer**

These layers progress from showing what the physical system is, to showing what is happening to it, and finally to exploring what could happen to it.

Importantly, verisimilitude is not simply about how realistic or detailed the graphics look. A photorealistic 3D model can still have low verisimilitude if it is only a static representation.

Instead, verisimilitude describes how closely the digital counterpart represents the physical system's information, current state, behavior, and potential future conditions.

## 1. Low Verisimilitude — Representative / Descriptive

### 1.1 Definition

A low-verisimilitude digital twin is a virtual representation of a real-world object, system, or environment that focuses primarily on structure, geometry, layout, and basic information.

It may use a 3D model, CAD model, or BIM model, but it has little or no live sensor-data integration.

**Main Question**

"What is the physical thing?"

### 1.2 Documented Example: Autodesk

Autodesk describes digital twins as digital representations that can connect information about physical assets and their environments.

At the descriptive level, a building can be represented through a detailed digital model containing information about rooms, walls, equipment, and other physical components.

The important characteristic for this level is that the model primarily represents the physical asset rather than continuously reproducing its changing operational state.

**Example link:** Autodesk — What Is a Digital Twin?

### 1.3 Houston-Based Illustrative Example: Minute Maid Park

A Houston example of how a low-verisimilitude digital twin could be used is Minute Maid Park.

A digital model could represent the stadium's seating sections, entrances, corridors, field dimensions, structural components, and other architectural features.

However, if the model does not receive live information, it would not know whether the retractable roof is currently open, how many people are inside the stadium, or what the current temperature is.

This demonstrates the difference between representing the structure of a place and representing its current operation.

### 1.4 Characteristics

**Provides:**

- Physical geometry and layout
- Structural information
- Design and construction information
- Visual representation of the physical asset

**Does not provide:**

- Continuous real-time sensor data
- Live operational conditions
- Predictive simulation

**Verisimilitude: Visual / Structural Replication**

## 2. Medium Verisimilitude — Monitoring / Connected

### 2.1 Definition

A medium-verisimilitude digital twin builds on a physical representation by connecting the virtual model to real-world operational information.

Sensors and Internet of Things (IoT) devices can provide data such as temperature, humidity, $CO_2$ levels, occupancy, energy consumption, proximity, or equipment status.

The digital twin can therefore reflect changes occurring in the physical environment.

**Main Question**

"What is happening right now?"

### 2.2 Documented Example: Charles H. Wright Museum

The Charles H. Wright Museum provides an example of a connected digital-twin application.

The museum developed a detailed 3D model of its facility and connected it with sensors that measure conditions such as $CO_2$, temperature, humidity, and proximity.

The sensor information can be displayed within the 3D model, allowing users to understand conditions in the physical museum.

This represents a higher level of verisimilitude than a static model because the digital representation is connected to the physical environment and receives operational data.

**Example link:** Charles H. Wright Museum Digital Twin — Autodesk

### 2.3 Houston-Based Illustrative Example: George R. Brown Convention Center

A Houston-based example of a medium-verisimilitude digital twin could be the George R. Brown Convention Center.

Instead of simply displaying floor plans and room layouts, a connected digital twin could incorporate live facility information such as air temperature, energy consumption, equipment status, and occupancy levels.

Building operators could use the digital model to determine current conditions in an exhibit hall or monitor energy usage throughout the facility.

The important difference from the Minute Maid Park example is that the digital twin would no longer represent only what the building looks like. It would also represent what is currently happening inside it.

### 2.4 Characteristics

**Provides:**

- A digital representation of the physical facility
- Live or near-real-time sensor information
- Environmental monitoring
- Operational awareness
- Information about changing conditions

**Generally lacks compared with a predictive twin:**

- Advanced predictive simulation
- Extensive "what-if" scenario testing
- Sophisticated forecasting of future behavior

**Verisimilitude: Operational Awareness**

## 3. High Verisimilitude — Predictive / Simulation

### 3.1 Definition

A high-verisimilitude digital twin goes beyond representing and monitoring a physical system.

It combines real-world data, computational models, simulation, and predictive analytics to approximate how the physical system behaves.

A high-verisimilitude digital twin can be used to test "what-if" scenarios, predict future conditions, evaluate potential changes, stress-test systems, and support decision-making before changes are made to the physical system.

**Main Question**

"What could happen?"

### 3.2 Documented Example: NASA National Airspace System

NASA's work involving the National Airspace System (NAS) provides an example of a predictive digital-twin application.

NASA uses digital-twin technology and simulation to study current and future airspace operations.

These simulations can incorporate air-traffic information with simulated aircraft and can operate using both fast-time and real-time simulation.

Researchers can use these environments to evaluate proposed changes and technologies without disrupting actual flight operations.

This represents a high level of verisimilitude because the goal extends beyond representing physical appearance or monitoring current conditions. The system can be used to explore how the larger system may behave under different conditions.

**Example link:** NASA — Aeronautics

### 3.3 Houston-Based Illustrative Example: Buffalo Bayou Flood Management

A Houston-based application of a high-verisimilitude digital twin could involve Buffalo Bayou and Houston-area flood management.

A predictive flood-management digital twin could combine physical terrain data, drainage networks, rainfall information, weather forecasts, and real-time streamflow measurements.

Computational models could then simulate severe storm scenarios and predict how floodwaters might move through different areas.

Emergency planners could use these simulations to evaluate potential flooding, determine where resources may be needed, and test possible response strategies before or during a severe weather event.

This illustrates the defining characteristic of a high-verisimilitude digital twin: it is not limited to telling users what is happening now. It can be used to explore what might happen next.

### 3.4 Characteristics

**Provides:**

- Real-world operational data
- Computational and physics-based models
- Predictive analysis
- Simulation
- "What-if" scenario testing
- Stress-testing
- Evaluation of potential future conditions
- Support for planning and decision-making

**Verisimilitude: Behavioral Intelligence and Prediction**

## 4. Comparison of the Three Levels

| Level | Layer | Main Question | What the Digital Twin Does | Example |
|---|---|---|---|---|
| **Low** | Representative | What is it? | Represents physical structure, geometry, and information | Autodesk / Minute Maid Park |
| **Medium** | Monitoring | What is happening? | Connects the digital model to current operational and sensor data | Charles H. Wright Museum / George R. Brown Convention Center |
| **High** | Predictive | What could happen? | Uses data, computational models, and simulation to predict and test future scenarios | NASA NAS / Houston flood-management application |

## 5. The Progression of Verisimilitude

The three levels can be summarized as an increasing connection between the physical system and its digital counterpart.

**LEVEL 1 — REPRESENTATION**

Physical System → Digital Representation

*What is it?*

**LEVEL 2 — MONITORING**

Physical System → Sensors / IoT → Digital Representation

*What is happening?*

**LEVEL 3 — PREDICTION**

Physical System ↔ Real-World Data ↔ Computational Model

↓ Simulation / Prediction

*What could happen?*

## 6. Key Takeaway

The three levels of verisimilitude demonstrate how digital twins can become increasingly connected to their physical counterparts.

At the low level, the digital twin primarily represents the form and structure of a physical system. It answers, “What is it?”

At the medium level, the digital twin incorporates real-world data and monitors the current state of the physical system. It answers, “What is happening?”

At the high level, the digital twin combines real-world data with computational models and simulations to investigate future behavior and possible scenarios. It answers, "What could happen?"

Therefore, verisimilitude is not determined by graphical quality alone. Instead, it measures how deeply the digital representation corresponds to the physical system's structure, current conditions, behavior, and potential future states.

**Representation → Monitoring → Prediction**

Visual / Structural Replication → Operational Awareness → Behavioral Intelligence and Prediction

## Team

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